Stat 215A Lecture Notes

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1 Week 7

1.1 Monday

1.2 Wednesday

Recall that Ω and $\mathcal{C}$ where $\mathcal{C}$ is a set of sets and U is a simply just a set.

Proposition. $\sigma(\mathcal{C} \cap U) = \sigma(\mathcal{C}) \cap U$

Proof. ($\supseteq$) Define the set

$$\mathcal{G} = \{A : A \cap U \in \sigma(\mathcal{C} \cap U)\}.$$

Last time we showed that

(1) $\Omega \in \mathcal{G}$

(2) for every $A \in \mathcal{G}$, $A^c \in \mathcal{G}$.

Now we will show that for a sequence of sets $A_n \in \mathcal{G}$ for all $n \in \mathbb{N}$, their union is contained in the set. Let A_n where $n \in \mathbb{N}$ is a sequence of sets in $\mathcal{G}$. Our goal is to show that

$$\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{G}$$

that is,

$$\bigcup_{n \in \mathbb{N}} A_n \cap U \in \sigma(\mathcal{C} \cap U).$$

Since $A_n \in \mathcal{G}$, we have

$$A_n \cap U \in \sigma(\mathcal{C} \cap U) \quad \forall n \in \mathbb{N}.$$

Since $\mathcal{C} \cap U$ is a sigma algebra, we have that

$$\bigcup_{n \in \mathbb{N}} (A_n \cap U) \in \sigma(\mathcal{C} \cap U).$$

But this tells us that

$$\bigcup_{n \in \mathbb{N}} A_n \cap U \in \sigma(\mathcal{C} \cap U) \implies \bigcup_{n \in \mathbb{N}} A_n \cup U \in \sigma(\mathcal{C} \cap U).$$

Hence, $\bigcup_{n \in \mathbb{N}} A_n \in \sigma(\mathcal{C} \cap U)$. ■

Proposition. $\sigma(\mathcal{C}) \cap U \subseteq \sigma(\mathcal{C} \cap U)$.

Proof. Note that $\mathcal{C} \subseteq \mathcal{G}$. Indeed, if $A \in \mathcal{C}$, then $A \cap U \in \mathcal{C} \cap U \subseteq \sigma(\mathcal{C} \cap U)$. Also, $\sigma(\mathcal{C}) \subseteq \mathcal{G}$ because $\mathcal{C} \subseteq G$, we have

$$\sigma(\mathcal{C}) \subseteq \sigma(G) = \mathcal{G}.$$

To show finally that $\sigma(\mathcal{C}) \cap U \subseteq \sigma(\mathcal{C} \cap U)$. Let $B \in \sigma(\mathcal{C}) \cap U$. Hence, $B = A \cap U$ for some $A \in \sigma\mathcal{C}$. Since $\sigma(\mathcal{C}) \subseteq \mathcal{G}$, $A \cap U \in \sigma(\mathcal{C} \cap U)$. Hence, this tells us that $B \in \sigma(\mathcal{C} \cap U)$. Since B was arbitrary,

$$\sigma(\mathcal{C}) \cap U \subseteq \sigma(\mathcal{C} \cap U).$$

■

Example. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{C} = \{\{1, 3, 5\}, \{2, 4, 6\}\}$ and $u = \{1, 2\}$. Our goal is to show that

$$\sigma(\mathcal{C} \cap u) = \sigma(\mathcal{C}) \cap u.$$

Indeed, notice that

(1) $C_1 \cap u = \{1, 3, 5\} \cap \{1, 2\} = \{1\}$;

(2) $C_2 \cap u = \{2, 4, 6\} \cap \{1, 2\} = \{2\}$.

Then we have

$$\underbrace{\sigma(\mathcal{C} \cap U)}_{\sigma-\text{algebra on } u} = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

and

$$\underbrace{\sigma(\mathcal{C})}_{\sigma\text{-algebra on } \Omega} = \{\emptyset, \{1, 3, 5\}, \{2, 4, 6\}, \{1, 2, 3, 4, 5, 6\}\}$$

and

$$\sigma(\mathcal{C}) \cap u = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

Hence, we see that

$$\sigma(\mathcal{C} \cap u) = \sigma(\mathcal{C}) \cap u.$$

Recall the proof of prop E.1.5 says in its last paragraph

"If $A \in \sigma(\mathcal{C}) \implies A \in \mathcal{G}$"

This implies that $A \cap U \in \sigma(\mathcal{C} \cap U)$ and so $A \in \sigma(\mathcal{C} \cap U)$. It follows that $\sigma(\mathcal{C}) \cap U \subseteq \sigma(\mathcal{C} \cap U)$.

Example (Simple Counter-Example to the Author's Proof)**.** Let $\Omega = \{1, 2, 3\}$ and $\mathcal{C} = \{\{1\}, \{2\}\}$ and $U = \{2, 3\}$. Then

$$\sigma(\mathcal{C}) = \{\emptyset, \Omega, \{1\}, \{2\}, \{2, 3\}, \{1, 3\}, \{3\}, \{1, 2\}, \Omega\}$$

This is a $\sigma-$algebra on Ω. Also,

$$\sigma(\mathcal{C} \cap U) = \sigma(\{\{\emptyset, \{2\}\}\}) = \{\emptyset, \{2\}, \{3\}, \{2, 3\}\}.$$

let $A = \{1\} \in \sigma(\mathcal{C})$. Furthermore, $A \cap U \in \sigma(\mathcal{C} \cap U)$ since $A \cap U = \emptyset$. But note that $A = \{1\} \notin \sigma(\mathcal{C} \cap U)$.

1.3 The Measure Function

Definition (Measure Function)**.** Let $\mathcal{F}$ be an algebra. A **measure function** $\mu : \mathcal{F} \to (0, \infty) \cup \{\infty\}$ that satisfies

(1) (**Empty set**) $\mu(\emptyset) = 0$

(2) (**Countable Additivity**) If $E_n n \in \mathbb{N}$ is a sequence of pairwise disjoint events in $\mathcal{F}$, then

$$\mu\Big(\bigcup_{n\in\mathbb{N}} E_n\Big) = \sum_{n=1}^{\infty} \mu(E_n).$$

(3) If $\mathcal{F}$ is a $\sigma-$algebra, the triplet $(\Omega, \mathcal{F}, \mu)$ is a measure space.

Definition (Probability Measure). A **Probability measure** is a measure that satisfies $\mu(\Omega) = 1$.

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Proposition (Continuity of a Measure). Any measure $\mu(\Omega) < \infty$ satisfies the following properties:

(i) (Finite additivity): For a finite sequence of disjoint sets $E_1, E_2, \dots, E_k$

$$\mu\Big(\sum_{i=1}^{k} E_i\Big) = \sum_{i=1}^{k} \mu(E_i).$$

Proof. Suppose we have an $\mathcal{F}-$algebra where $\mu : \mathcal{F} \to [0, \infty) \cup \{\infty\}$. If E_n where $n \in \mathbb{N}$ is a sequence of disjoint sets; that is, $E_i \cap E_j = \emptyset$ if $i \neq j$. Then

$$\mu\Big(\bigcup_{i=1}^{\infty} E_i\Big) = \sum_{i=1}^{\infty} \mu(E_i).$$

Note that in (i), given the above, we can write

$$\mu\Big(\bigcup_{i=1}^{k} E_i\Big) = \mu\Big(\sum_{i=1}^{k} E_i\Big) = \sum_{i=1}^{k} \mu(E_i).$$

This statement can be easily proven using property (1) of the definition of a measure where all indices i past k, we can set $E_i = \emptyset$. Indeed, we will consider three cases:

(1) $i \neq j$ where $i, j \in \{1, 2, \dots, k\}$ implies $E_i \cap E_j = \emptyset$ where $E_1, E_2, \dots, E_k$ are all disjoint.

(2) $i \neq j$ where $i, j \in \{k+1, k+2, \dots\}$ implies $E_i \cap E_j = \emptyset \cap \emptyset = \emptyset$.

(3) $i \in \{1, 2, \dots, k\}$ and $j \in \{k+1, k+2, \dots\}$ implies $E_i \cap E_j = E_i \cap \emptyset = \emptyset$.

From (i)-(iii), we have E_i where $i \in \mathbb{N}$ is a disjoint sequence of sets. The above proposition includes E_n for $n \in \mathbb{N}$ disjoint and

$$\mu\Big(\bigcup_{i=1}^{\infty} E_n\Big) = \sum_{i=1}^{\infty} \mu(E_n),$$

which implied that $\bigcup_{i=1}^{\infty} E_i \in \mathcal{F}$. Note that because

$$\begin{aligned}\mu\Big(\bigcup_{i=1}^{\infty} E_i\Big) &= mu\Big(\bigcup_{i=1}^{\infty} E_i\Big) = \sum_{i=1}^{\infty} \mu(E_i) \\ &= \sum_{i=1}^{k} \mu(E_i) + \sum_{i=k+1}^{\infty} \mu(\emptyset) \\ &= \sum_{i=1}^{k} \mu(E_i) + 0 \\ &= \sum_{i=1}^{k} \mu(E_i).\end{aligned}$$

■

Recall that if f is continuous, then $x_n \to x$ which implies that $f(x_n) \to f(x)$.

Proposition (Measure is Continuous from Below)**.** If E_n is a non-decreasing sequence to E, $E \in \mathcal{F}$ and $E_n \in \mathcal{F}$ for all $n \in \mathbb{N}$, then $\mu(E_n)$ converges to $\mu(E)$ where $\mu(E) \in \{\sup E, \infty\}$.